MASTER IN ASTROPHYSICS AND SPACE SCIENCE

Nicolaus Copernicus Astronomical Center of the Polish Academy of Sciences

Numerical Simulations of AGN Accretion Disks in PLUTO

CAMK PAN Summer Student Program

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Overview

Numerical Setup
- Black Hole Adaptation
- Radiation Physics
- Compton Coupling

Disk Classifier
- Passive Tracer
- Construction
- Validation

Results
- Relaxation
- Steady State
- Density Waves

Discussion
- Limitations
- Outlook
- Conclusions

Black Hole Adaptation

- Adapted Viscous Star Disk Setup from Numerical Atlas to Nonrotating SMBH
  - Solved 2.5D on Spherical Grid in r and θ with Axial & Equatorial Symmetry
  - Replaced Stellar Surface with Absorbing Diode Inner Boundary
  - Replaced Newtonian with Paczyński–Wiita Potential to Reproduce ISCO

$$\Phi = -\frac{GM}{r - r_s}$$

  - Measured Lengths and Times in Gravitational Units with c for Velocities

$$r_g = GM/c^2 \qquad t_g = r_g/c$$

  - Embedded Kluźniak–Kita Disk from $R = 6$ with $\epsilon = 0.1$ in Hydrostatic Corona
  - Reinjected Analytic Disk at Outer Boundary and Blocked Coronal Inflow

Radiation Physics

- Enabled Nonrelativistic Two Moment Radiation Module with Full Speed of Light
  - Fixed Physical Units by Black Hole Mass of $1 \times 10^8\ M_\odot$ and Density Unit

$$r_g \approx 1.48 \times 10^{13}\ \mathrm{cm} \qquad t_g \approx 493\ \mathrm{s} \qquad \rho_0 = 5.8 \times 10^{-10}\ \mathrm{g\,cm^{-3}}$$

  - Balanced Initial Viscous Heating and Radiative Cooling at $\alpha = 0.5$
  - Included Kramers Absorption as well as Thomson Scattering Opacities
- Added Effective Compton Opacity since PLUTO Treats Scattering as Coherent

$$\kappa_C = \kappa_{\mathrm{es}} \frac{4 k_B T_r^4}{m_e c^2 \left(T_g + T_r\right)\left(T_g^2 + T_r^2\right)}$$

  - Reproduces Compton Exchange Rate and Prevents Gas Heating up to 1×10^{11} K
  - Rate Capped above 1×10^9 K, Decoupled above $0.9\,c$ and Gas Floor at 1×10^4 K

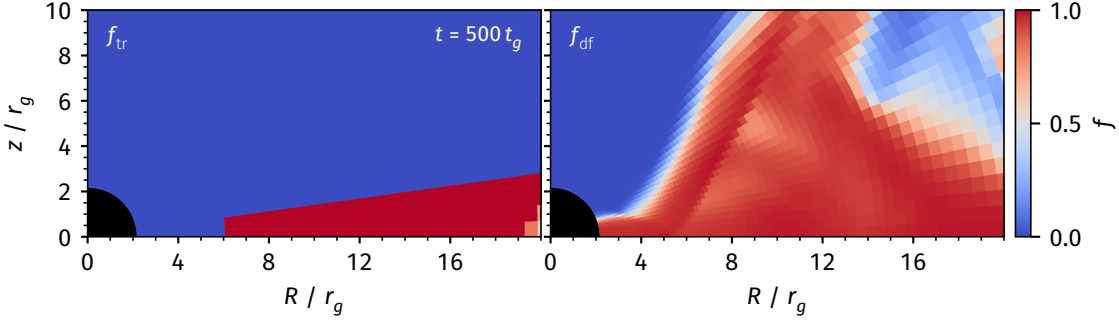
Passive Tracer

- Advected Tracer Switches Viscosity, Resistivity, Cooling and Opacities to Disk

$$\partial_t q + \mathbf{v} \cdot \nabla q = 0$$

  - Stays at Initial Wedge Once Radiation Enabled while Disk Evolves
  - Cause Not Isolated in Code, Only Restricted Physics Affected
- Replaced Tracer by Classifier Built from Current State of Each Cell Alone
  - Same Disk Condition as Initial Setup, Identical to Tracer at Start
  - Robust to Restarts by Writing Classifier State with Every Output

Radiation Enabled – Tracer & Classifier



Construction

- Classified Cells by `DiskFraction` $f \in [0, 1]$ from Density and Rotation
  - Stored Disk and Corona Reference Densities in 90 Logarithmic Radial Bins
  - Located Cell Density between References on Logarithmic Scale

$$s = \frac{\log \rho - \log \rho_c}{\max\left(\log \rho_d - \log \rho_c, 0.5\right)}$$

  - Multiplied Density and Keplerian Rotation Criteria with Logistic Function

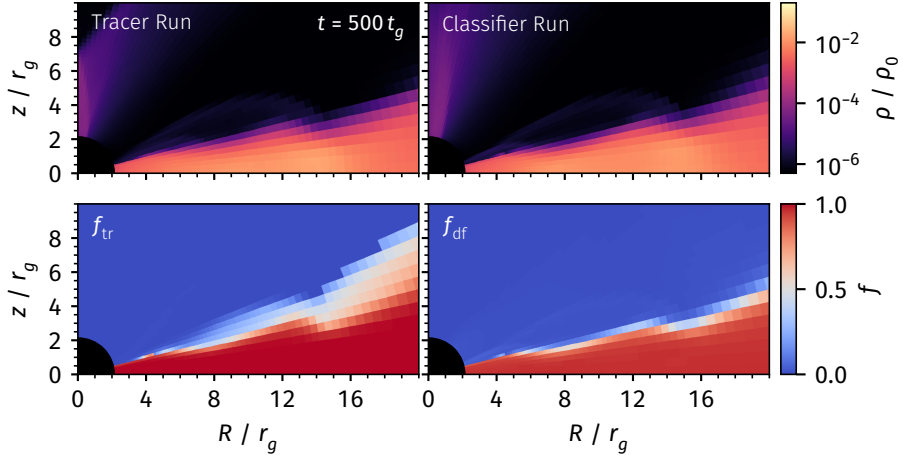
$$f = \sigma\left(\frac{s - 1/6}{0.3}\right)\sigma\left(\frac{|v_\phi|\sqrt{R} - 0.5}{0.1}\right)$$

  - Updated References as f and $1 - f$ Weighted Means via Moving Average with Weight $w = 1 - \exp(-\Delta t/\tau)$ and Time Constant $\tau = 6\, t_g$

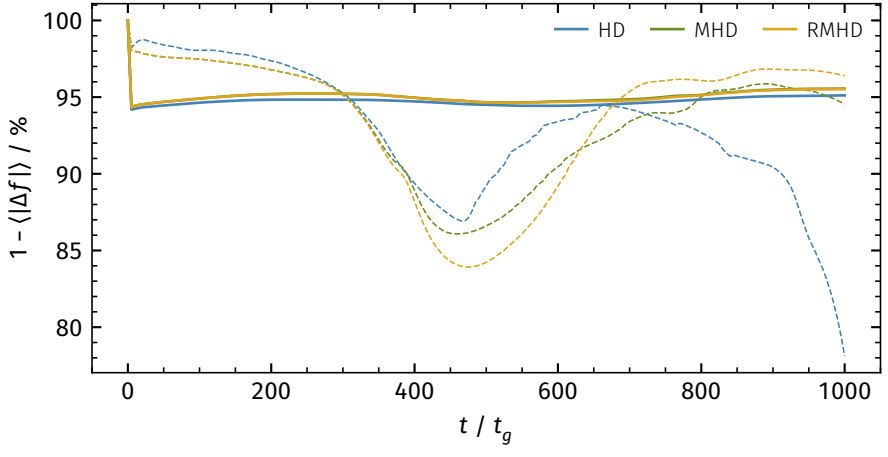
Versions & Guards

- First Version Used Fixed Weight per Step and Inverted after Disk Disruption
  - Disk Reference Ten Times below Corona Reference at Radii from 2.4 to 3.4
  - Inversion Sustained Itself since References Built from Classification
  - Plunging Gas inside ISCO Rotates Faster than Keplerian and Counted as Disk
- Current Version Repairs Itself instead of Locking Error in Place
  - Enforced Minimum Span of 0.5 dex between Disk and Corona References
  - Copied References from Nearest Valid Bin, Searching Outward First
  - Smoothed with Time Constant for Same Memory at Every Resolution

Validation – Hydrodynamics



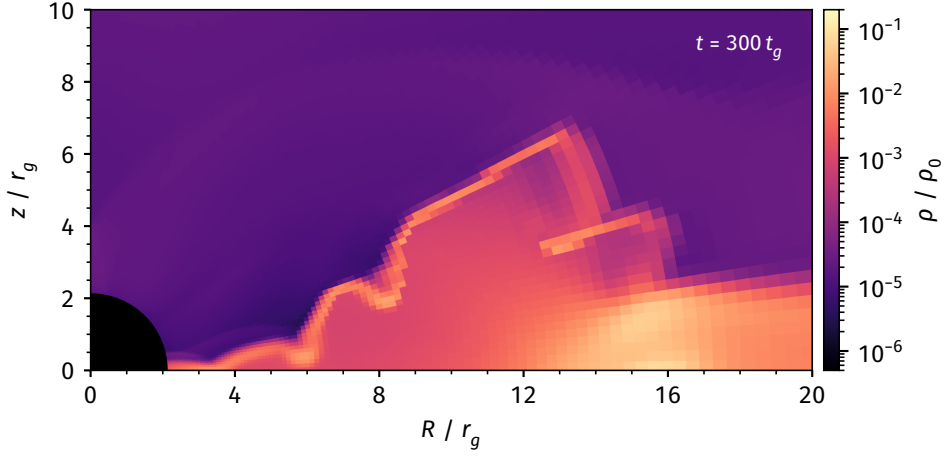
Validation – Agreement



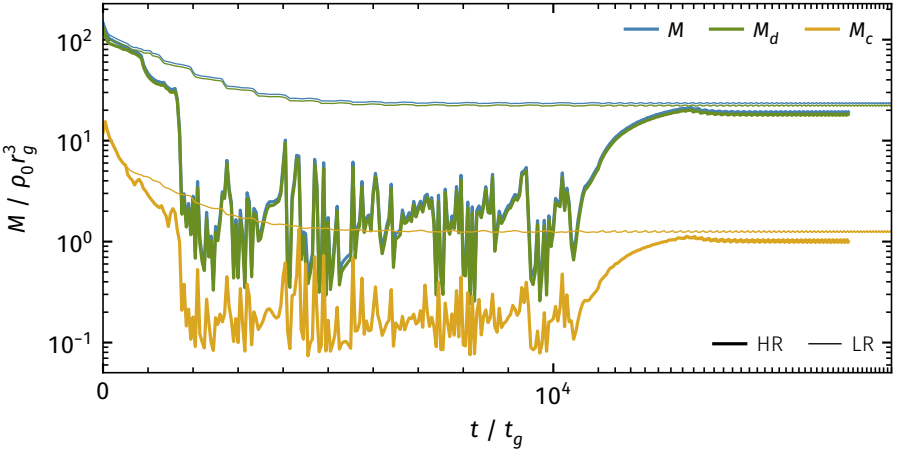
Radiation Runs

- Simulated Radiation Setup with Compton Coupling at Two Resolutions
  - LR with 90×45 Cells up to $t = 5 \times 10^4$ and First Classifier Version
  - HR with 120×90 Cells up to $t = 4 \times 10^4$ and Current Classifier Version
  - Output Every 50 Gravitational Times, Refined to 5 for Last 10 000
- Inflated Disk and Broke up Dense Surface Layer during First Hundreds of t_g
  - HR Disk Blown Apart and Rebuilt by $t \approx 15000$, LR Declined Smoothly
  - Corona Drained Early and Not Replenished due to Blocked Inflow
- Steady State when Disk Mass, Radiation Energy, Accretion Rate and Luminosity Remain within Two Standard Errors of Final Fifth
  - Reached at $t = 23725$ for HR and $t = 9225$ for LR

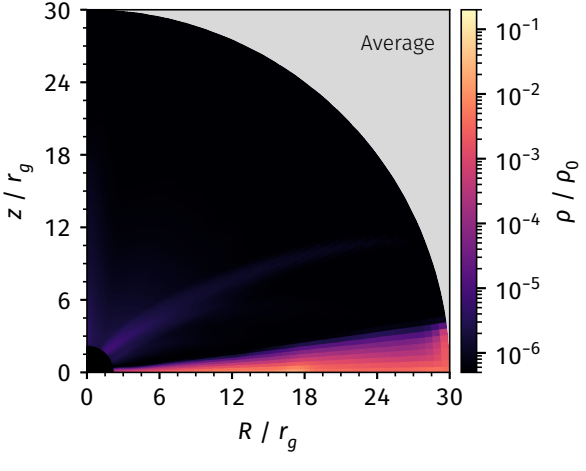
Early Evolution – Density



Relaxation – Mass



Steady State – Density



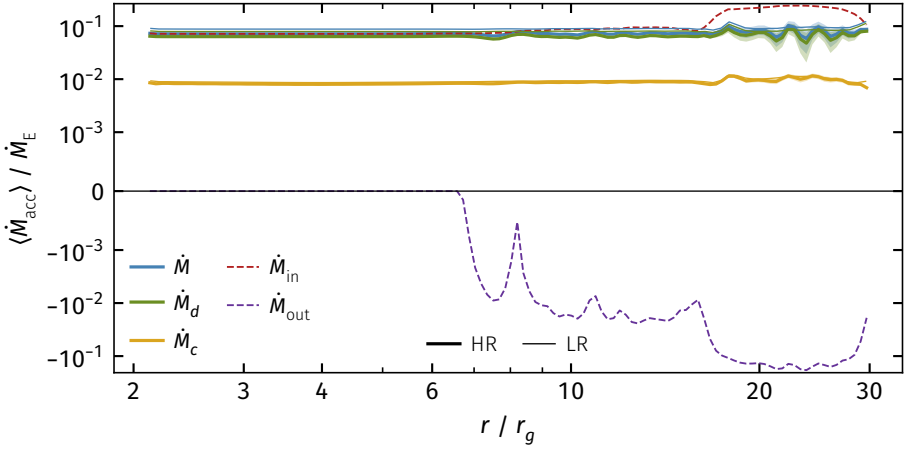
Steady State

- Thinner Disk with Aspect Ratio **0.057** in HR and **0.042** in LR at $R = 10$
- Inflow Equilibrium out to $r \approx 17$ with Strong Circulation Further Out
  - Accretion Rate **0.073** and **0.091** in Eddington Units, about **0.3** $M_\odot\,yr^{-1}$
  - Luminosity **0.056** and **0.070** with Radiative Efficiency $\eta \approx 0.048$
- Surface Density below Steady Thin Disk Profile

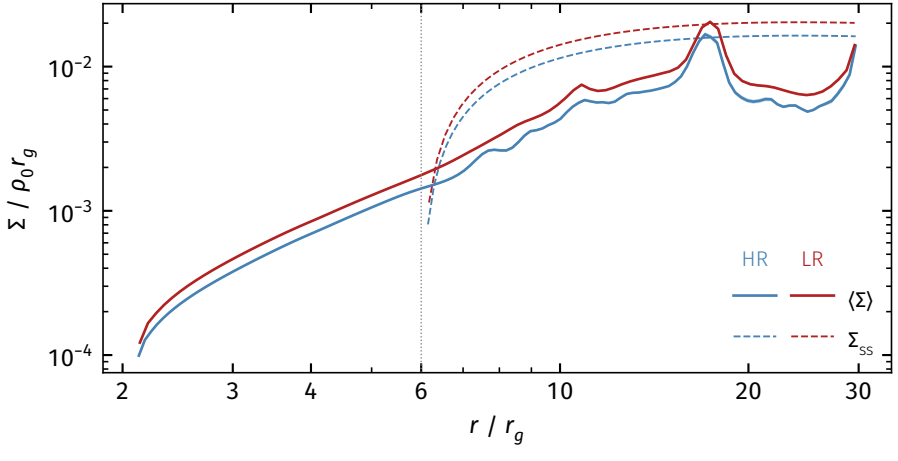
$$\Sigma_{SS} = \frac{\dot{M}}{3\pi\nu}\left(1 - \sqrt{6/R}\right)$$

  - Effective Viscosity Two to Three Times Higher than Prescribed α
- Viscous Heating Balanced by Compton Cooling, Ratio of **0.8** to **0.9**
  - Gas a Hundred to Two Thousand Times Hotter than Radiation at Midplane

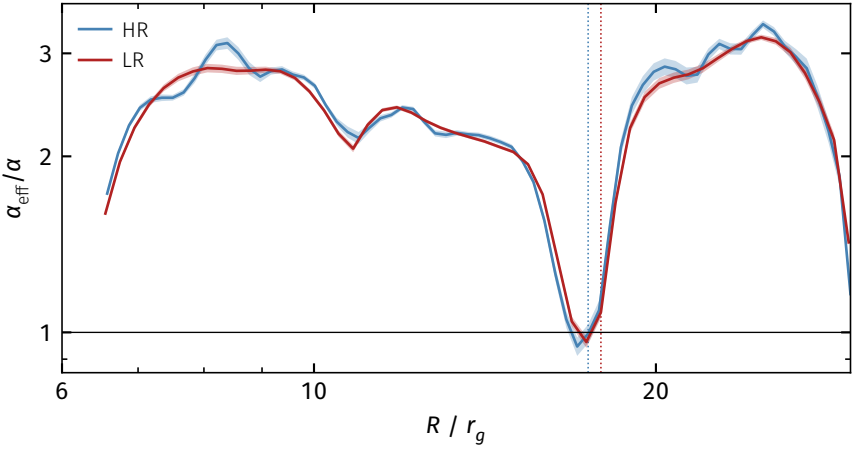
Steady State – Accretion



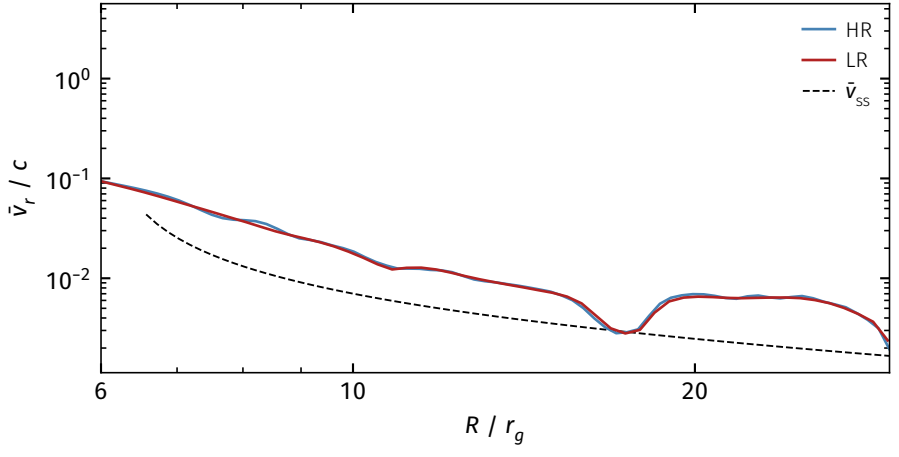
Steady State – Surface Density



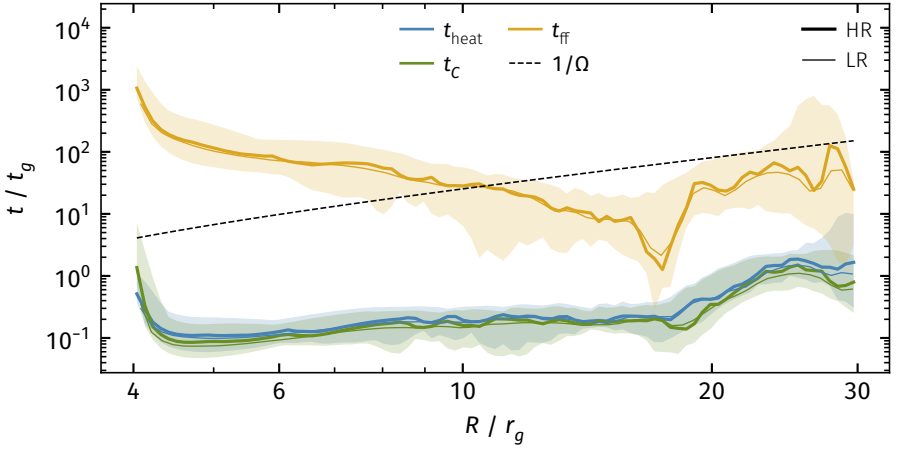
Steady State – Effective Viscosity



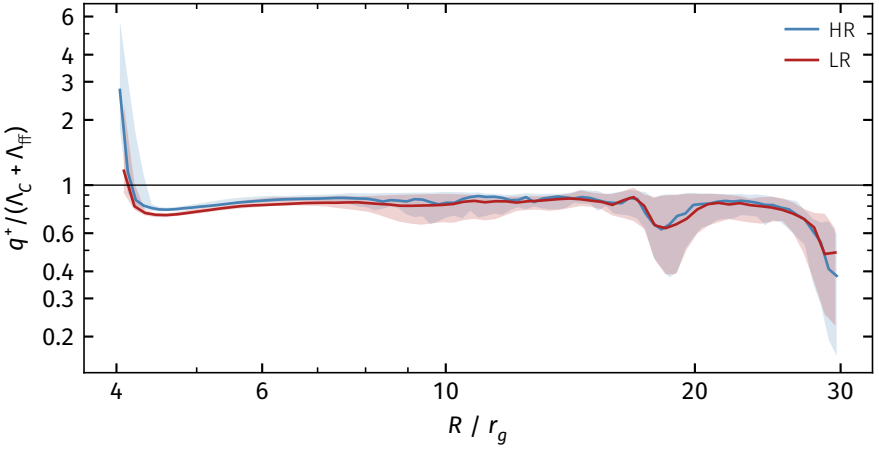
Steady State – Inflow Speed



Steady State – Heating & Cooling Times



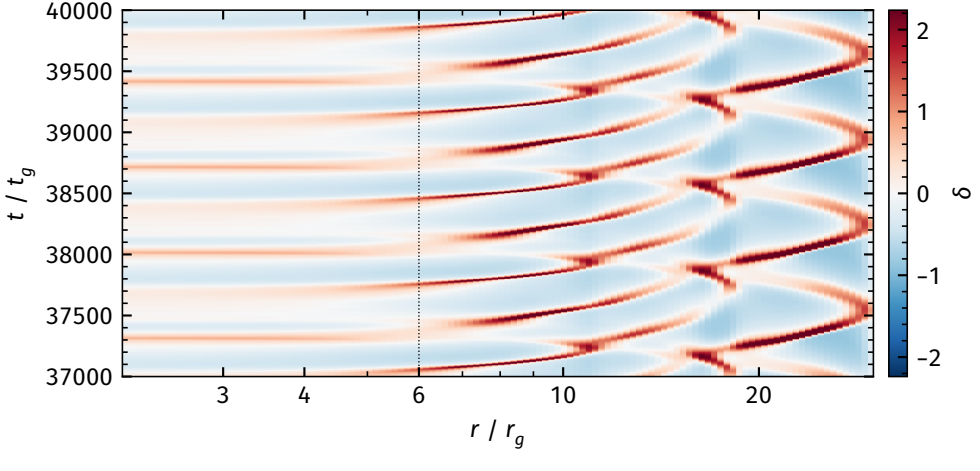
Steady State – Heating & Cooling Ratio



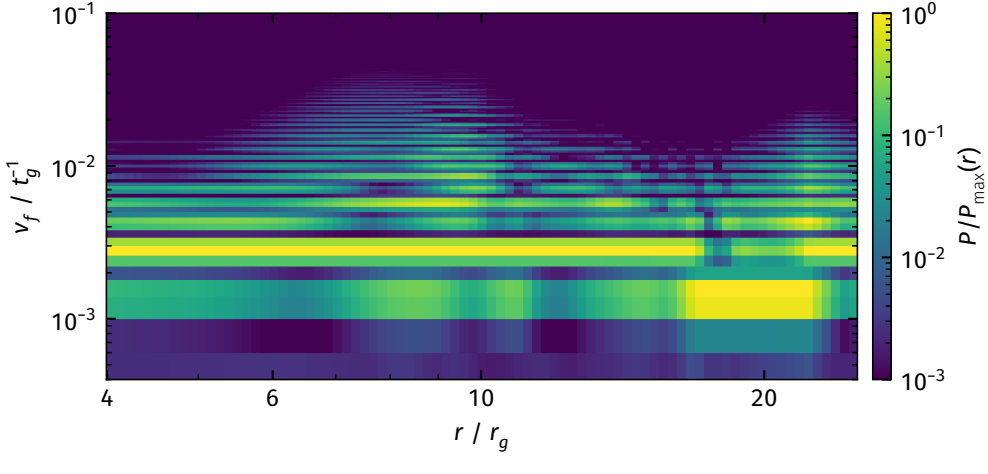
Density Waves

- Found Quasiperiodic Density Waves Traveling Outward in Both Runs
  - Starting near ISCO and Propagating to $R = 17$ and Beyond
  - Strongly Nonlinear with Relative Amplitude from **0.25** up to **0.8**
  - Period of **349** Gravitational Times, about Two Days for Our SMBH
- Phase Speed of $0.019\,c$ Steady between Radii of **4** and **17**
  - **2.2** Times Adiabatic Sound Speed with Radiation, so Not Plain Sound Waves
  - Same Period at Both Resolutions Suggests Origin in Setup rather than Numerics
  - Close to Thermal Time $1/(\alpha\Omega)$ of $330\,t_g$ at Outer Boundary

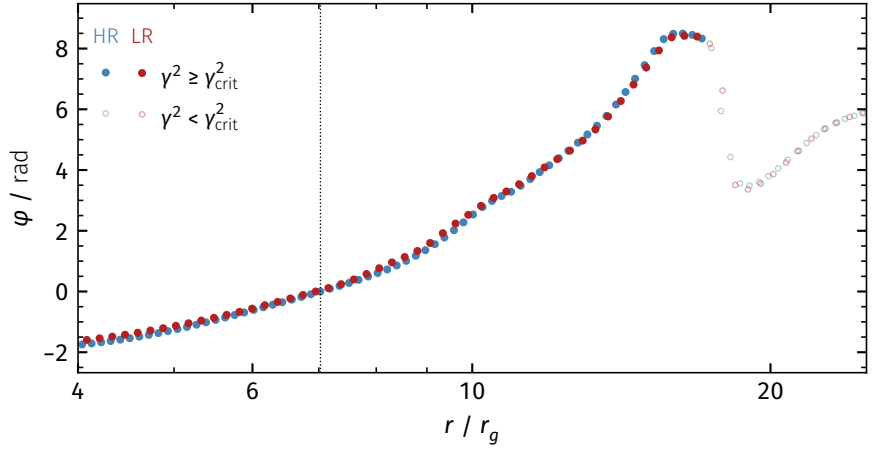
Density Waves – Perturbation



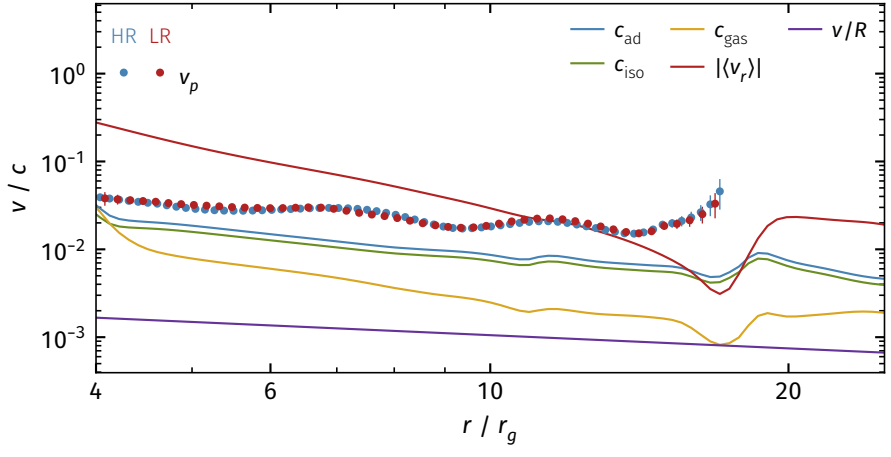
Density Waves – Power Spectrum



Density Waves – Phase



Density Waves – Phase Speed



Resolution Comparison

Quantity	HR	LR
Cells ($r \times \theta$)	120×90	90×45
Classifier Version	Current	First
Steady State Start / t_g	23 725	9225
Accretion Rate / $\dot{M}_E$	0.073	0.091
Luminosity / L_E	0.056	0.070
Efficiency	0.048	0.049
Equilibrium Radius / r_g	17.4	17.9
Aspect Ratio ($R = 10$)	0.057	0.042
Cells per Scale Height	3.3	1.2
Wave Period / t_g	349	349
Phase Speed / c	0.0190	0.0195

Limitations

- Viscosity Uses Sound Speed of Initial Disk, Fixed in Time
  - Removes Feedback Driving Thermal and Viscous Instability, so Stability No Test
- Boundaries Set Gas Supply and May Cause Inflow Stream and Wave Period
  - Mass Budget Open by 3 % between $r = 2.5$ and 25, 18 % for Full Domain
- Gas Temperatures Likely Too High for Setup to Be Fully Trusted
  - Corona up to 1×10^{11} K with $k_B T$ about 46 Times Electron Rest Energy
  - Neither Relativistic Compton Corrections nor Pair Production Included
- Plunging Gas inside about Four r_g Moves Superluminally in Pseudo Potential
- Axisymmetric Hydrodynamics Replaces MRI Turbulence by α Prescription

Outlook & Conclusions

- Next Steps for More Physical Setup
  - Derive Viscous Sound Speed from Evolving Total Pressure
  - Extend Coarse Domain to Move Outer Boundary Away from Region of Interest
  - Start from Standard Thin Disk Solution Suited to Radiation Pressure
- Built Working SMBH Radiation Setup in PLUTO with Effective Compton Coupling
  - `DiskFraction` Agrees with Tracer to **95 %** by Mass and Follows Disk with Radiation
  - Steady States at about a Tenth of Eddington Rate with Compton Balanced Heating
  - Nonlinear Density Waves with Period of about **350** t_g Traveling Outward

Thank You!

- A. Mignone et al., *PLUTO: A Numerical Code for Computational Astrophysics* (ApJS) 2007
- M. Čemeljić, *Atlas of Numerical Solutions for Star-Disk Magnetospheric Interaction* (A&A) 2019
- J. D. Melon Fuksman et al., *A Two-Moment Radiation Hydrodynamics Scheme Applicable to Simulations of Planet Formation in Circumstellar Disks* (ApJ) 2021
- B. Paczyński, P. J. Wiita, *Thick Accretion Disks and Supercritical Luminosities* (A&A) 1980
- Y.-F. Jiang et al., *Super-Eddington Accretion Disks around Supermassive Black Holes* (ApJ) 2019

Resources on GitHub: here

Model Equations

- Initial Disk Pressure and Density from Analytic Solution of Kluźniak & Kita

$$C = \frac{2}{5\epsilon^2}\left(\frac{1}{r} - \frac{1 - 5\epsilon^2/2}{R}\right) \qquad p = \epsilon^2 C^{5/2} \qquad \rho = C^{3/2}$$

- Corona in Hydrostatic Equilibrium with $\rho_c = 0.01$ in Code Units

$$\rho = \rho_c r^{-3/2} \qquad p = 0.4\rho_c r^{-5/2}$$

- Shear Viscosity with Sound Speed of Initial Disk, Restricted by Classifier

$$\eta_v = \frac{2}{3}\alpha\rho\epsilon^2 C R^{3/2} f \qquad v = \frac{2}{3}\alpha\epsilon^2 R^{1/2}$$

- Density Floor at 5×10^{-7} in Code Units with Constant Sound Speed and Momentum

Opacity Model

- Kramers Free Free and Bound Free Absorption with Guillotine Factor

$$\kappa_K = \left(K_{\rm ff} + K_{\rm bf}\right) \rho\, T_g^{-7/2} \qquad \kappa_{\rm es} = 0.2\,(1 + X)$$

- Compton Exchange Held at Cap above $T_{\rm max} = 1 \times 10^9$ K

$$\kappa_C = \kappa_{\rm es} \frac{4 k_B T_r^4 \left(T_{\rm max} - T_r\right)}{m_e c^2 \left(T_g^4 - T_r^4\right)}$$

- Gated by Classifier and Decoupled near Speed of Light, Thomson Scattering on All Gas

$$f_d = \max\left(\frac{f - 0.05}{0.95}, 0\right) \qquad g_v = \left[1 + \exp\left(\frac{\beta - 0.9}{0.03}\right)\right]^{-1}$$

$$\kappa_{\rm abs} = g_v f_d \left(\kappa_K + \kappa_C\right) \qquad \kappa_{\rm sc} = g_v \left(\kappa_{\rm es} - f_d \kappa_C\right)$$

Classifier Details

- Corona Reference as $1 - f$ and Disk Reference as f Weighted Mean Density per Bin
  - Disk Estimate Counts Only above One Thousandth of Bin Volume
  - Bins without Valid Estimate Copy Nearest Valid Bin
- Guards Added after Inversion in First Version
  - Span below 0.5 dex Copies Both References from Nearest Sufficient Bin
  - Disk Reference Raised to Minimum Span if No Such Bin Exists
  - Fixed Weight 0.05 per Step Replaced by Time Constant τ
- Optional Geometric Mode Independent of Classification
  - Disk Reference from Maximum Density, Corona from Geometric Mean above $z/R = 0.5$

Analysis Methods

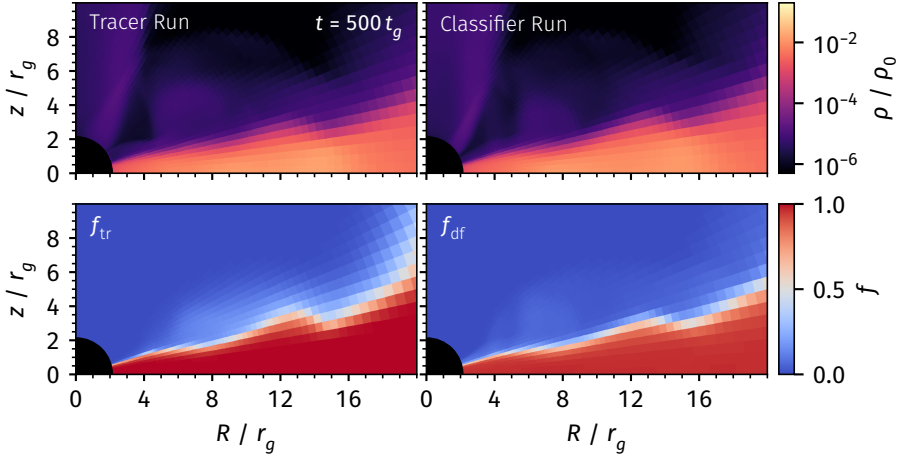
- Standard Errors from Block Averages over **1000** Gravitational Times
- Relaxation from Running Means over $W = 2500$ Compared with Final Fifth

$$D(t_c) = \frac{|\bar{X}_W(t_c) - X_{\mathrm{ref}}|}{\sigma}$$

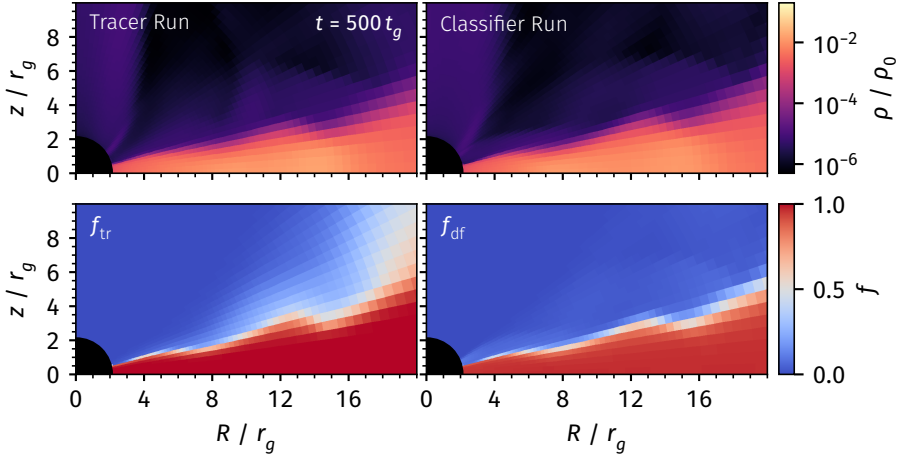
- Inflow Equilibrium where Accretion Rate within **20 %** of Inner Value
- Welch Spectra of Surface Density Perturbation $\delta = \Sigma/\langle\Sigma\rangle - 1$
  - Phase at Peak Frequency Relative to $r = 7$ for Significant Coherence Only
  - Local Phase Speed from Linear Fits over **15 %** in Radius

$$v_p = \omega \,/\, \frac{\mathrm{d}\varphi}{\mathrm{d}r}$$

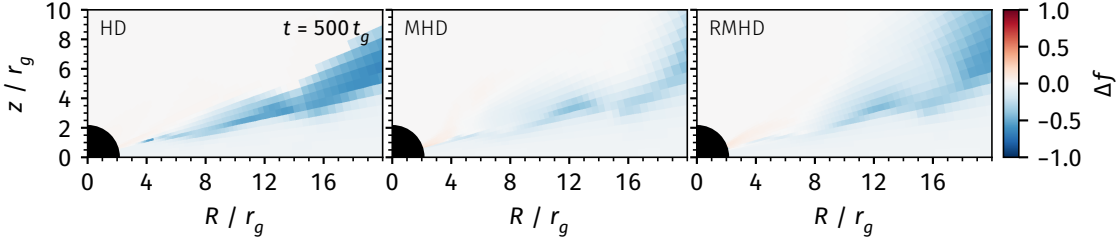
Validation – Magnetohydrodynamics



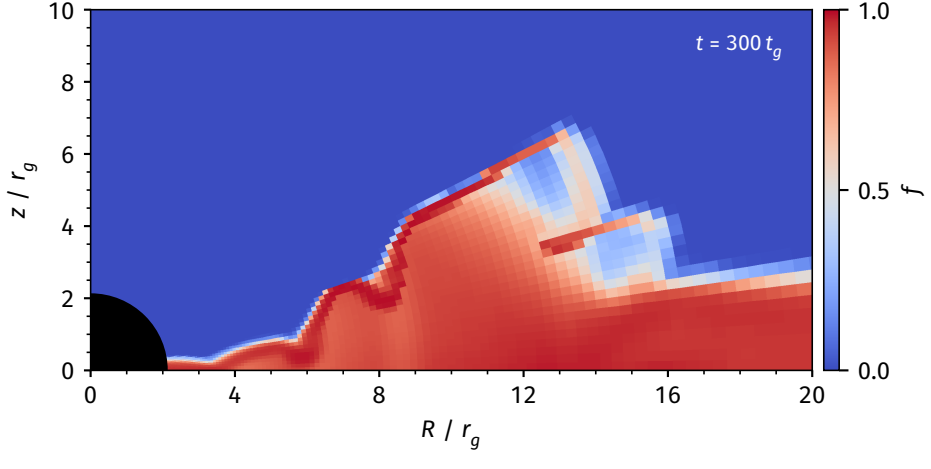
Validation – Resistive Magnetohydrodynamics



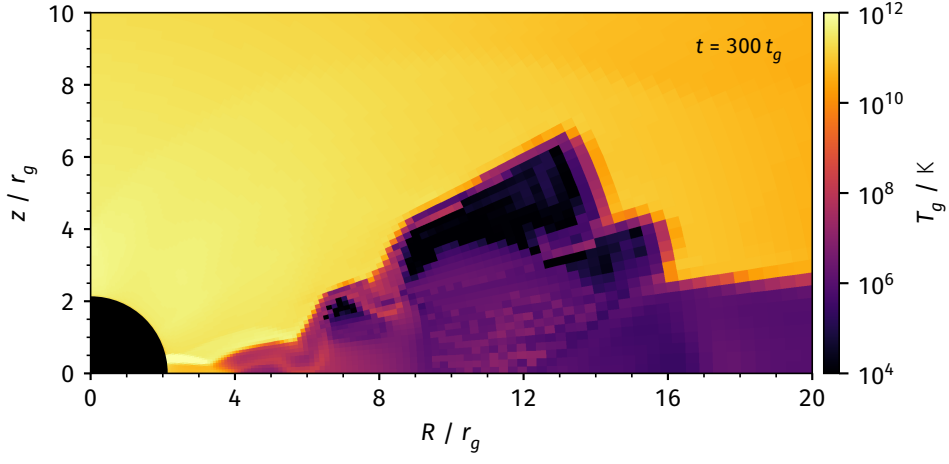
Validation – Classifier Minus Tracer



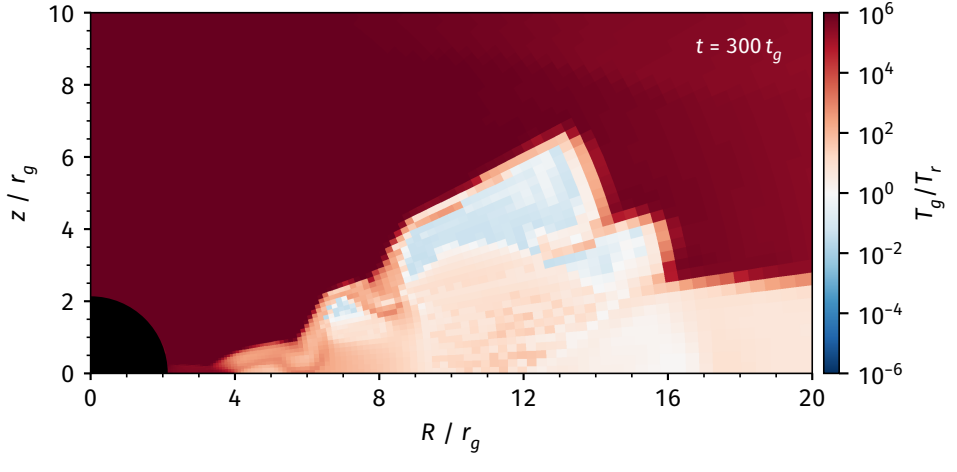
Early Evolution – DiskFraction



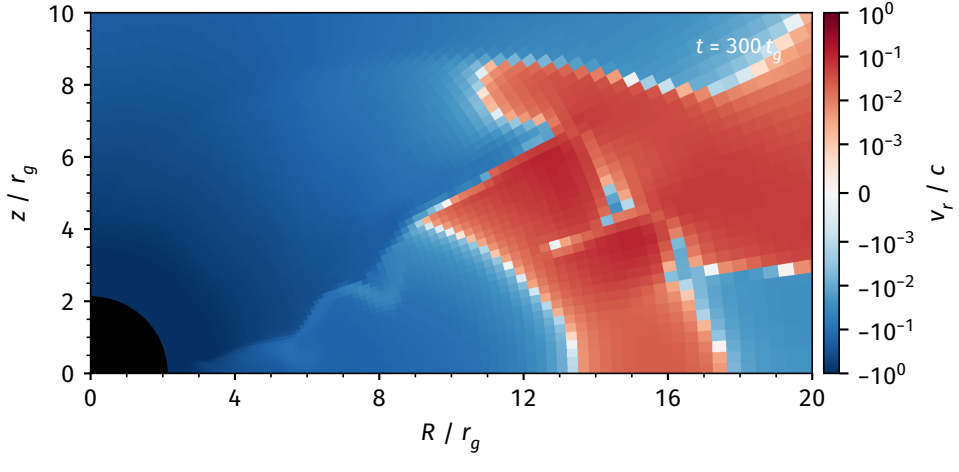
Early Evolution – Gas Temperature



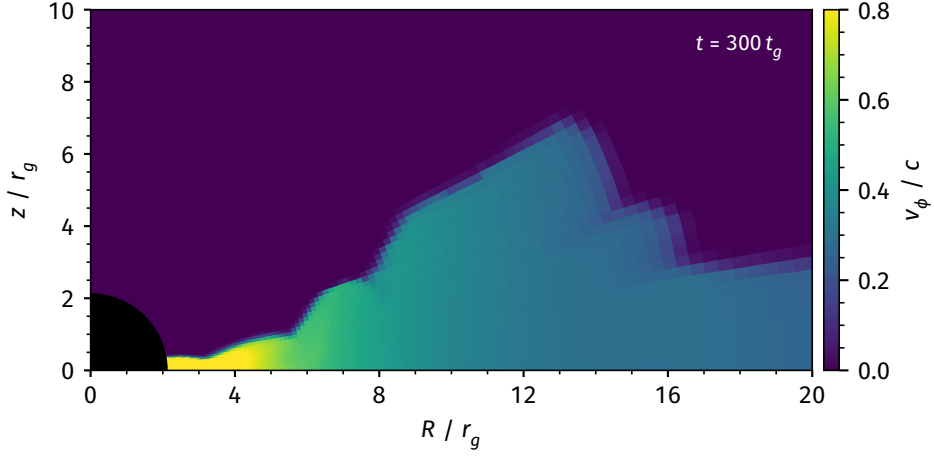
Early Evolution – Temperature Ratio



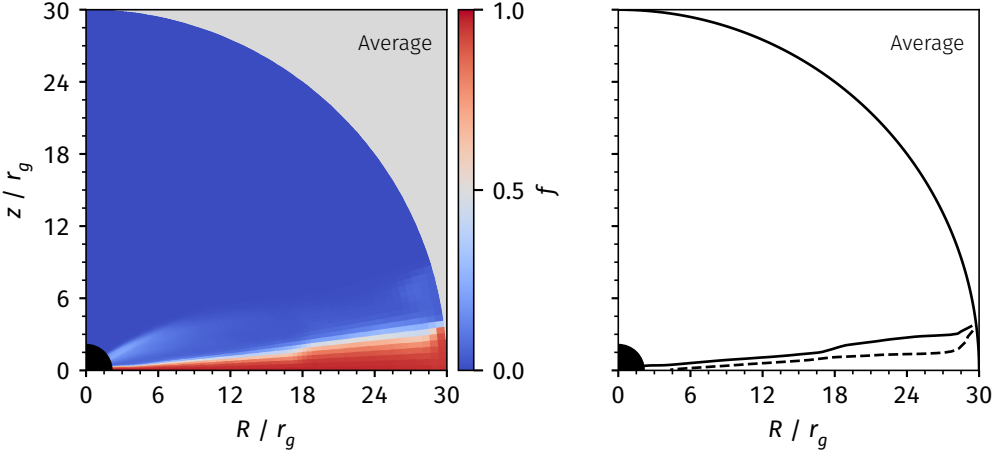
Early Evolution – Radial Velocity



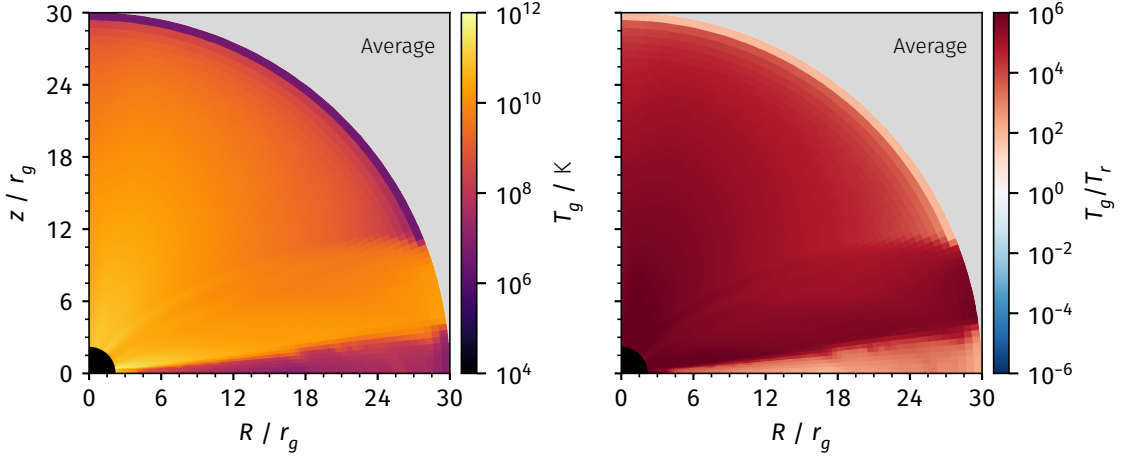
Early Evolution – Rotation Velocity



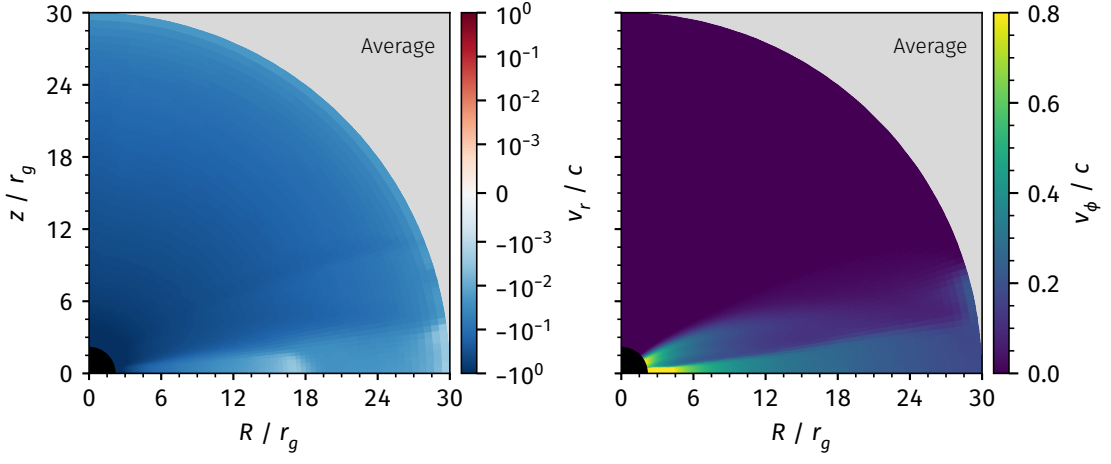
Steady State – DiskFraction & Disk Surface



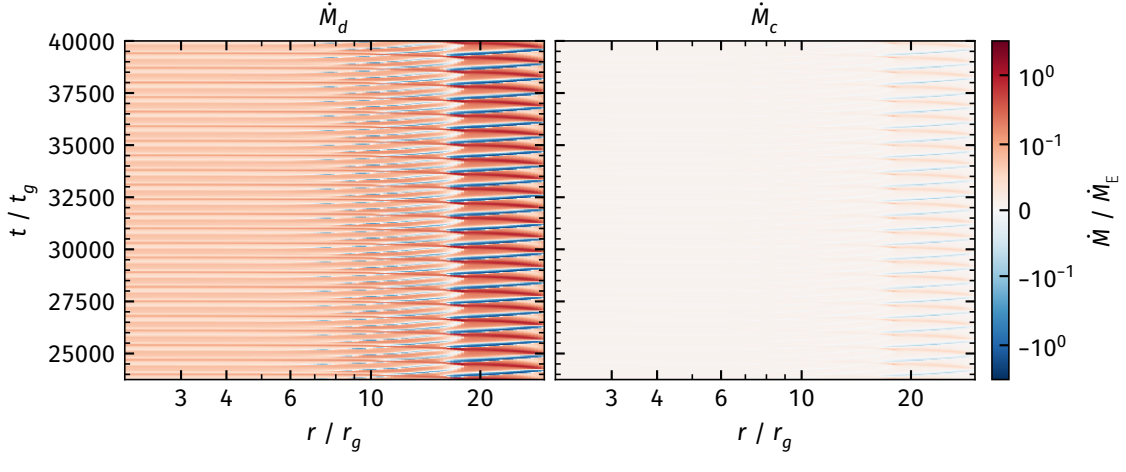
Steady State – Temperatures



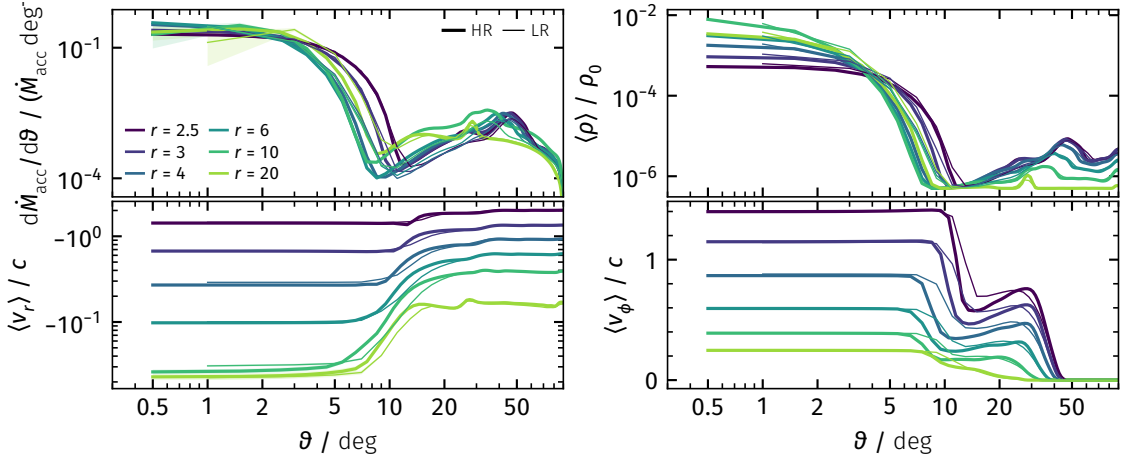
Steady State – Velocities



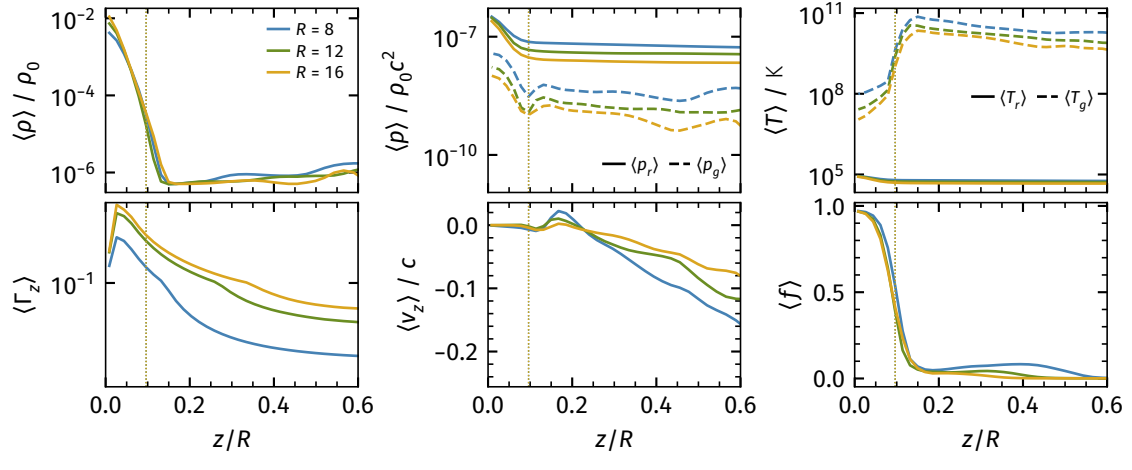
Steady State – Accretion Rate Map



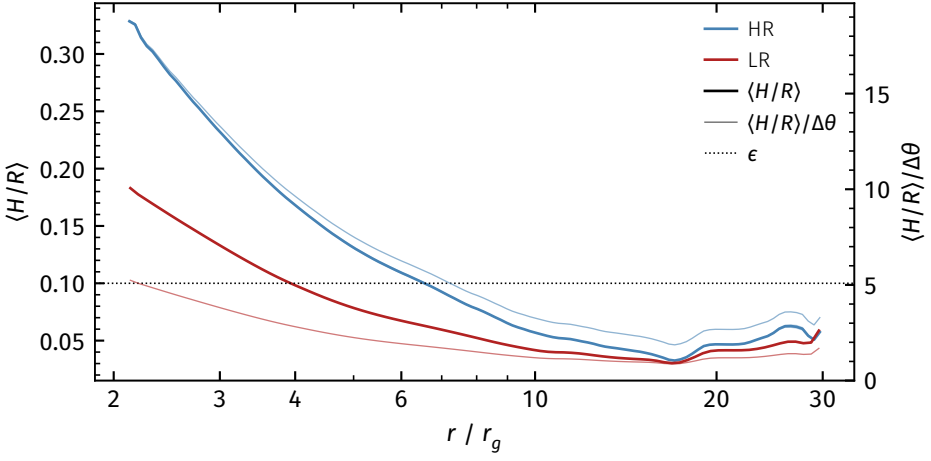
Steady State – Angular Profiles



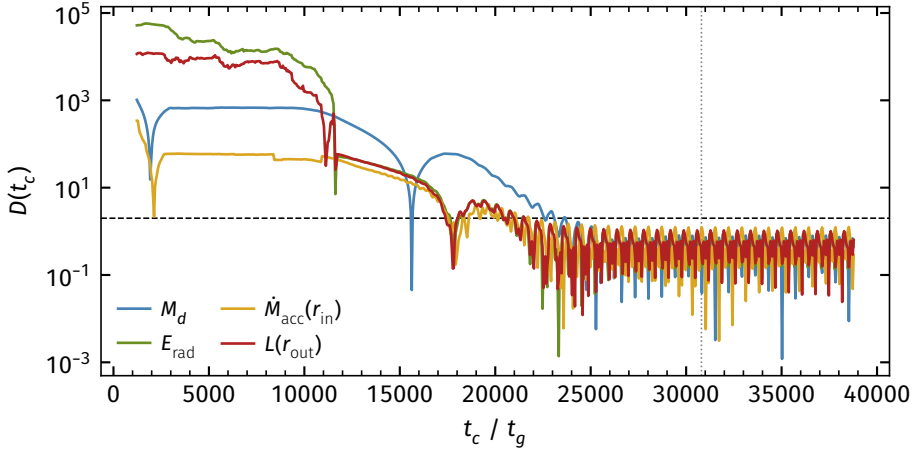
Steady State – Vertical Profiles



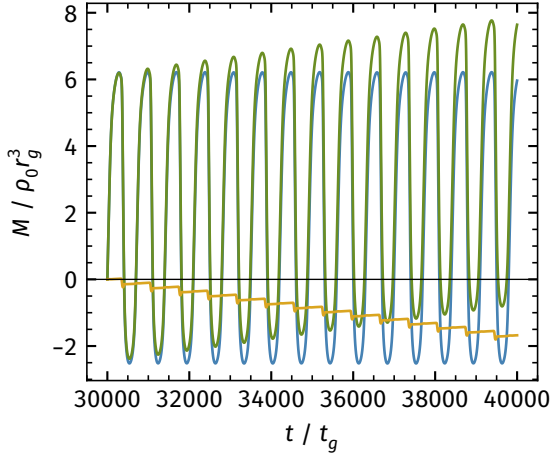
Steady State – Aspect Ratio

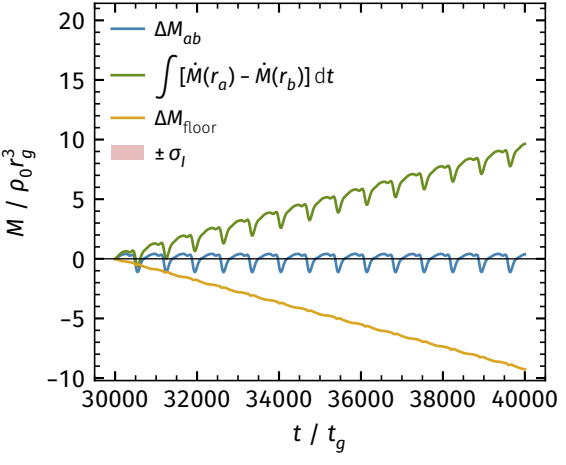


Diagnostics – Relaxation Statistic

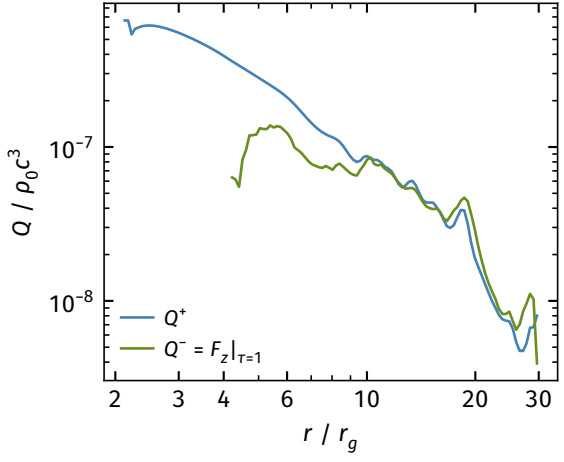


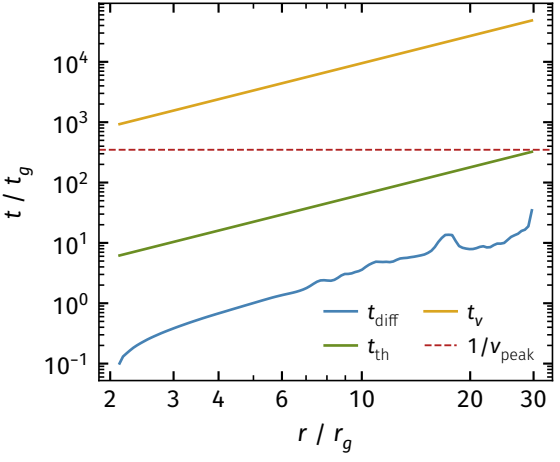
Diagnostics – Mass Budget



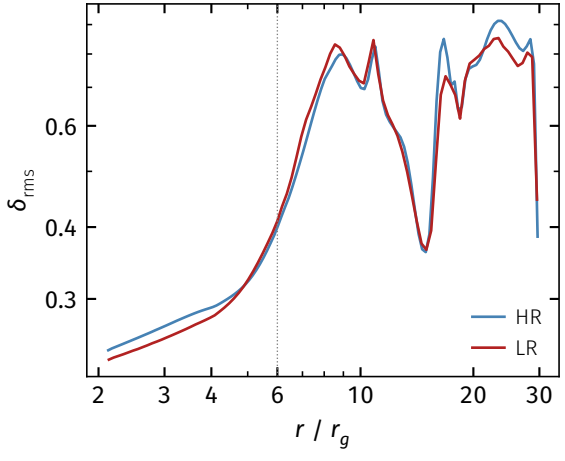


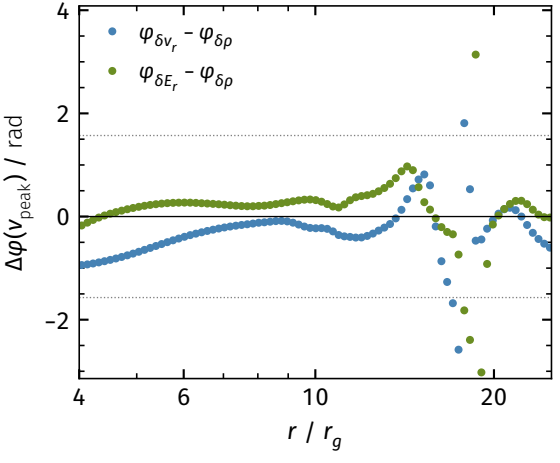
Diagnostics – Energy Balance





Diagnostics – Wave Amplitude & Cross Phase





Diagnostics – Timescales

